

Sepsis Recognition and Early Treatment Protocol

Sunrise Medical Center

Document Title: Sepsis Recognition and Early Treatment Protocol

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Policy Owner: Sepsis Program Coordinator, Denise Falk, RN

Distribution: All Sunrise Medical Center Staff

1. Purpose

Sepsis is a leading cause of preventable in-hospital mortality. This protocol establishes the screening tools, escalation pathway, and time-bound treatment bundle used at Sunrise Medical Center to identify sepsis early and initiate treatment before irreversible organ damage occurs.

2. Screening

All patients are screened for sepsis at triage and with every set of vital signs using the modified quick Sequential Organ Failure Assessment (qSOFA) combined with the hospital's electronic sepsis alert, which fires automatically in the EHR when two or more criteria are met.

- Respiratory rate ≥ 22 breaths per minute
- Altered mentation (GCS < 15)
- Systolic blood pressure ≤ 100 mmHg
- Temperature $> 38.3^{\circ}\text{C}$ or $< 36^{\circ}\text{C}$ combined with suspected infection source

3. The Sepsis Hour-1 Bundle

Once sepsis is suspected, the following bundle must be initiated within one hour of recognition, not one hour of ED arrival. The nurse who identifies the alert is empowered to begin bundle elements under the standing order set without waiting for physician co-signature.

- 1 Measure serum lactate level; repeat within 2–4 hours if initial lactate is elevated above 2 mmol/L.
- 2 Obtain blood cultures (two sets, from separate sites) prior to antibiotic administration whenever feasible, without delaying antibiotics beyond the one-hour window.
- 3 Administer broad-spectrum antibiotics per the Sepsis Empiric Antibiotic Guide (SMC-PHARM-018), tailored to suspected source.
- 4 Begin rapid administration of 30 mL/kg crystalloid fluid for hypotension or lactate ≥ 4 mmol/L.
- 5 Apply vasopressors if the patient remains hypotensive during or after fluid resuscitation to maintain mean arterial pressure ≥ 65 mmHg.

4. Source Control

Identification and control of the infectious source — including abscess drainage, catheter removal, or surgical consultation — must occur as soon as clinically feasible and always within 12 hours of diagnosis. Delays in source control are the most common preventable cause of treatment failure reviewed in mortality committee cases.

5. Reassessment

Patients are reassessed at 3 hours and 6 hours from bundle initiation, with repeat lactate, vital signs, and a focused volume status exam. Patients not responding to initial resuscitation are escalated to the ICU team for evaluation of central venous access and vasopressor titration.

6. Documentation and Core Measure Reporting

Every suspected sepsis case is logged on the Sepsis Bundle Compliance worksheet, which captures every bundle-element timestamp. This data supports CMS SEP-1 core measure reporting and is reviewed weekly by the Sepsis Program Coordinator, with non-compliant cases escalated for individual case review.

7. Roles and Responsibilities

Role	Responsibility
Triage/Bedside RN	qSOFA screening, alert response, bundle initiation under standing orders
Emergency Physician	Source identification, antibiotic selection confirmation, escalation decisions
Pharmacist	Antibiotic verification and rapid dispensing, dosing adjustment for renal function
Charge Nurse	Bed placement prioritization for ICU-level escalation
Sepsis Coordinator	Weekly compliance audit and case review coordination

8. Related Documents

SMC-PHARM-018 Sepsis Empiric Antibiotic Guide; SMC-LAB-006 STAT Lactate Turnaround Standard; SMC-CC-009 ICU Escalation Criteria; SMC-QI-014 CMS SEP-1 Core Measure Reporting Procedure.

Appendix A: Frequently Asked Questions

Q: Does the sepsis bundle clock start at triage or at physician evaluation? A: The clock starts at the moment sepsis is clinically suspected or the electronic alert fires, whichever occurs first — not at physician bedside evaluation, which may be delayed.

Q: What if blood cultures cannot be drawn before antibiotics due to difficult access? A: Antibiotics must not be delayed beyond the one-hour target waiting for line access; document the attempt and draw cultures as soon as access is obtained, ideally within a few minutes of first antibiotic dose.

Q: Can the fluid bolus be given to a patient with heart failure? A: Yes, but with closer monitoring; use smaller, more frequent boluses with reassessment of respiratory status between each, and involve the physician early if there is concern for volume overload.

Q: What lactate level triggers repeat measurement? A: Any initial lactate above 2 mmol/L requires a repeat level within 2 to 4 hours to assess trend and response to treatment.

Appendix B: Common Scenarios

Scenario — Elderly patient with atypical presentation: An elderly nursing home resident presents with confusion and low blood pressure but no fever. This still meets sepsis alert criteria and the full bundle should be initiated; atypical presentations are common in geriatric sepsis.

Scenario — Suspected sepsis in a patient already on the medical floor: The bedside nurse activates the same electronic alert and standing order set used in the ED; the rapid response team may be called to assist with bundle initiation on the inpatient unit.

Revision History

Version	Date	Description
1.0	Sep 15, 2022	Initial release
1.1	Apr 4, 2025	Updated lactate reassessment interval and vasopressor target